
CYBERSECURITY FOR PROFESSIONALS BOOTCAMP

TEAM-BASED LAB

Triaging SIEM Alerts and Simulating Incident Response

Week 7, Days 1–2 • Day 1 Individual • Day 2 Team

No software to install • No accounts to create • No cost

1. Before You Start

This lab runs on paper and in a spreadsheet. There is no system to log into, no console to learn, and no password to request. Everything you need is printed in this document.

That is deliberate. Tier 1 triage is a judgement skill, not a tool skill. You will learn the SIEM console itself in Week 8. Today is about whether you can look at an alert and make a call you can defend.

1.1 What you need

Item	Detail
A spreadsheet	Excel, Google Sheets or LibreOffice Calc. Your triage log lives here.
A document editor	For your escalation notes and, on Day 2, your after-action report.
A calculator	Your phone is fine. You will do distance and time arithmetic on the impossible-travel alerts.
This document	Keep it open. Sections 4 and 5 are reference material you will use constantly.

There are no trick questions, but there are hard ones

Twenty alerts are in the queue. Roughly half should be closed and roughly half escalated, which is close to what a real Tier 1 shift looks like.

Two of them are genuinely ambiguous. Reasonable analysts could disagree. Those are not mistakes in the lab - they are the point of it, and how you handle them is worth more marks than the easy calls.

Several alerts look alarming and are benign. Several look routine and are not. Read the reference data in Section 4 before you judge anything.

1.2 Scope and ethics

Read this before you start

Every alert, log line, email address, account and transaction in this document is fabricated for the exercise. Meridian Trust Bank does not exist.

The external addresses and domains in the queue belong to the scenario. Do not visit them, look them up in a service that connects to them, or paste them into any site that performs live lookups.

Do not run any scanning or lookup tool against anything, anywhere, as part of this lab. There is nothing here that requires it.

2. The Scenario

You are a Tier 1 security analyst in Meridian Trust Bank's Security Operations Centre. It is Wednesday morning. The overnight queue has not been worked because the night analyst was pulled onto a separate incident, so twenty alerts are waiting for you.

Your shift lead wants the queue cleared by end of day, with a decision and a justification on every single alert.

2.1 How a tiered SOC works

Tier	What they do	What they hand on
Tier 1 - Monitoring and Triage	Works the alert queue. Applies runbooks. Decides close or escalate. Does not investigate deeply and does not take remediation action.	An escalation note to Tier 2, or a documented closure.
Tier 2 - Investigation	Takes escalations and investigates properly: pulls more data, correlates across systems, confirms or disproves the Tier 1 suspicion.	A confirmed incident to Tier 3, or a closed investigation.
Tier 3 - Hunt and Incident Response	Runs full incident response, hunts for related activity, coordinates containment and remediation.	A resolved incident and an after-action report.

The two ways a Tier 1 analyst fails

Escalating everything. If you escalate all twenty alerts, Tier 2 drowns and stops trusting your judgement. The queue exists precisely so that somebody filters it.

Closing something real. A single wrongly closed alert can be the one that mattered, and nobody will ever know it was wrong until the incident surfaces weeks later.

The job is holding both of those in mind at once. Your justification is what protects you either way - a closure with sound recorded reasoning is defensible even when it later turns out to be wrong.

3. How to Work Through This Lab

3.1 Time plan

Session	Section	What you are doing	Guide time
Day 1	4	Read the reference data before you touch the queue	30 minutes
Day 1	5	Read the four runbooks	45 minutes
Day 1	6	Work the twenty-alert queue	3 to 4 hours
Day 1	7	Write your escalation notes	1 hour
Day 2	8	Team kickoff and role assignment	30 minutes
Day 2	9	Investigation	2 hours
Day 2	10	Containment and remediation	1 to 2 hours
Day 2	11	Team after-action report	1 to 2 hours

3.2 Set up your triage log first

Create a spreadsheet with these columns before you read a single alert. Working the queue without a log is how analysts lose track of what they have already decided.

Column	What goes in it
Alert ID	MTB-A01 through MTB-D05
Type	Failed login, impossible travel, BEC indicator, or wire transfer
Decision	CLOSE or ESCALATE
Confidence	High, medium or low. Be honest.
Justification	Why. One or two sentences citing the specific detail that decided it.
Runbook step	Which runbook criterion you applied
Linked alerts	Any other alert in the queue you think is related

The Linked alerts column is the one people skip

Alerts arrive one at a time, in separate categories, often hours apart. Attacks do not.

Fill that column in as you go. If you finish the queue and it is completely empty, go back and look again.

3.3 What a good justification looks like

Standard	Example
Not acceptable	Looks fine. Closing.
Passing	Closing. The failed logins came from the user's own corporate IP and succeeded on the eighth attempt, which looks like a forgotten password.
Strong	Closing with high confidence. Seven failures then a success for cwalsh, all from 10.50.1.31, which the directory in Section 4.2 confirms is her assigned workstation. The sequence ran 08:41 to 08:44 on a Tuesday, inside business hours. The quarterly password rotation completed the previous day per Section 4.5, which makes a forgotten password the most likely explanation. Failed Login Runbook closure criterion 2 applies. No escalation, but I have noted it in case the same account appears again this week.

The pattern to copy

State the decision and your confidence. Name the specific evidence and where you got it. Say what ordinary explanation fits. Cite the runbook criterion. Note anything that would change your mind.

Five sentences. Every alert in your log should read like that.

4. Reference Data

Read this section before you work the queue. Most of the alerts cannot be decided correctly without it, and several look completely different once you have.

4.1 Meridian Trust at a glance

Item	Detail
Headquarters	Pittsburgh, Pennsylvania. All offices are in the Pittsburgh metro area.
Business hours	07:00 to 19:00 Eastern, Monday to Friday. Treasury Operations runs 07:00 to 17:00.
Wire cut-off	17:00 Eastern. Wires initiated after cut-off queue for the next business day.
Dual approval	Any outbound wire over 25,000 US dollars requires a second approver.
Today	Wednesday. The queue covers Monday night through this morning.

4.2 Staff directory (extract)

Account	Name and role	Assigned workstation	Notes
cwalsh	Carla Walsh, Retail Operations	10.50.1.31	-
dkoenig	Daniel Koenig, Infrastructure, domain admin	10.50.1.52	On approved leave all week
mnguyen	Minh Nguyen, Branch Support	10.50.1.44	New phone enrolled in MDM on Monday
rpatel	Rhea Patel, Credit Risk	10.50.1.60	-
jhaddad	Joseph Haddad, Commercial Lending	10.50.1.63	Approved travel, Miami conference, Tue to Fri
lbrennan	Liam Brennan, Data Services	10.50.1.71	Remote worker, uses corporate VPN daily
tokafor	Thomas Okafor, Treasury Operations	10.50.1.58	Authorised wire initiator
sgrigsby	Sara Grigsby, Audit	10.50.1.66	-
bfooster	Brian Foster, Marketing	10.50.1.80	-
dkessler	Diane Kessler, Chief Financial Officer	10.50.1.12	-
rharper	Ray Harper, Accounts Payable	10.50.1.49	Authorised payee administrator

4.3 Known network ranges

Range or address	What it is
10.50.1.0/24	Corporate workstation subnet, Pittsburgh offices
10.50.2.0/24	Server subnet
198.51.100.0/24	Meridian Trust corporate VPN egress. Physically located in a Columbus, Ohio datacentre. All remote-worker traffic appears to originate here.

Range or address	What it is
203.0.113.14	Meridian Trust mail gateway outbound address

The VPN egress range is the single most useful fact in this section

A remote worker sitting in their kitchen in Pittsburgh will appear to log in from Columbus, Ohio, because that is where the VPN traffic leaves Meridian Trust's network.

At least one alert in the queue looks like impossible travel purely because of this and is completely benign. Check every geolocation against this range before you escalate it.

4.4 Service accounts

Account	Purpose	Expected source	Expected behaviour
svc-backup-agent	Nightly backup job	10.50.2.25	Authenticates at 02:00 daily
svc-sql	Database service account	10.50.2.25	Authenticates continuously during business hours
svc-monitor	Estate monitoring agent	10.50.2.30	Checks in every 5 minutes

4.5 Recent change record

Date	Change	Reference
Monday	Quarterly password rotation completed for all Retail Operations staff	CHG-8841
Monday	Minh Nguyen issued a replacement phone, enrolled in MDM	CHG-8847
Two weeks ago	Backup agent credential rotated. Application team did not update the job.	CHG-8802, still open
Last Thursday	Payment of 250,000 US dollars to Harrow Logistics approved by CFO and Treasury lead	CR-2291

4.6 Approved wire beneficiaries (extract)

Beneficiary	Bank	First used	Typical amount
ADP Payroll Services	Wells Fargo, US	2019	128,000 to 132,000 US dollars, fortnightly
Harrow Logistics Ltd	Barclays, UK	2021	40,000 to 260,000 US dollars, irregular
Castlereagh Supply Co	PNC Bank, US	2022	80,000 to 95,000 US dollars, monthly
Delacroix Systems	Santander, ES	2023	15,000 to 22,000 US dollars, monthly

4.7 Travel time reference

You will need this for the impossible-travel alerts. Work out the distance, work out how fast the person would have had to move, and compare it against reality.

Mode	Realistic speed
Commercial airliner, cruise	500 to 560 miles per hour
Commercial flight, door to door	Add roughly 2 hours for airport time either side
Car on a motorway	65 to 75 miles per hour
Anything faster than about 600 miles per hour	Not physically possible for a person

Route from Pittsburgh	Great-circle distance
Columbus, Ohio	162 miles
Toronto, Canada	224 miles
Miami, Florida	1,014 miles
Bucharest, Romania	4,998 miles
Lagos, Nigeria	5,578 miles

5. The Runbooks

A runbook is the SOC's documented procedure for one alert type. It exists so that every analyst handles the same alert the same way, and so that a decision can be reviewed later against what the procedure actually said.

Apply them. If you close or escalate against a runbook you must say why, and that is a legitimate thing to do - runbooks do not cover everything. What is not acceptable is ignoring them silently.

5.1 Runbook: Failed Login

What the alert means

One or more authentication failures crossed a threshold. Thresholds are deliberately low, so most of these are benign.

What to check, in order

1. How many failures, over what period, and did any attempt eventually succeed?
2. Is the source address the user's own assigned workstation, the corporate VPN range, or somewhere else entirely?
3. One account or many? Failures spread across many accounts from one source is a different attack entirely.
4. Is the account privileged, a service account, or an ordinary user?
5. Does the timing fit the user's normal working pattern, and is there a change record that explains it?

Close if

- Fewer than 10 failures on a single ordinary account from that user's own workstation or the VPN range, followed by a success, inside business hours.
- A change record explains it, such as a password rotation or a new device enrolment.
- A service account failing from its expected source in its expected pattern, with a known open ticket.

Escalate if

- Failures span multiple accounts from a single source. This is password spraying and the count does not need to be high.
- The account is privileged and the attempt succeeded.
- The source is external or unrecognised.
- The activity falls outside the user's working pattern with no change record, and especially if the user is known to be away.

5.2 Runbook: Impossible Travel

What the alert means

The same account authenticated from two locations too far apart for the person to have travelled between them in the time available. The alert is generated automatically from IP geolocation, which is frequently wrong.

What to check, in order

1. Do the arithmetic yourself. Distance divided by elapsed time gives required speed. Compare against Section 4.7.
2. Is either address in the corporate VPN range? If so the geolocation is an artefact, not a movement.
3. Is there approved travel on record for this user?
4. What did the account do after the second login? A login is only half the story.
5. Was multi-factor authentication satisfied, and how - an approved push, or repeated prompts?

Close if

- The required speed is achievable by commercial flight and there is approved travel on record.
- One of the two addresses belongs to the corporate VPN egress range.
- The two locations are close enough that ordinary ground travel explains the gap.

Escalate if

- The required speed exceeds roughly 600 miles per hour. No person moves that fast.
- The second location is somewhere the user has never logged in from and there is no travel record.
- The account performed sensitive actions immediately after the second login.

5.3 Runbook: Business Email Compromise Indicator

What the alert means

Something in the mail environment matched a pattern associated with business email compromise. BEC is not malware. It is an attacker using a real or convincing mailbox to get money moved, and the technical footprint is very small.

What to check, in order

1. If a mailbox rule was created, what does it actually do? Rules that hide or delete messages are the single strongest BEC indicator there is.
2. Who created the rule, from what address, and at what time?
3. If a domain is involved, compare it character by character against the real one.

4. Does the display name match a real employee while the underlying address does not?
5. Did the message reach anyone, or was it blocked at the gateway?

Close if

- The activity is an ordinary mail feature used ordinarily, such as an out-of-office reply set by a user who is genuinely on leave.
- The message was blocked at the gateway, reached nobody, and the pattern is generic rather than targeted.

Escalate if

- A rule moves, deletes or marks as read any message matching financial keywords. Treat this as compromise until proven otherwise.
- A rule forwards mail to an address outside the bank.
- A lookalike domain is involved, whether or not the message was delivered.
- An executive or an authorised payment initiator is impersonated or targeted.

5.4 Runbook: Suspicious Wire Transfer

What the alert means

An outbound transfer matched a fraud-monitoring rule. Meridian Trust moves large sums legitimately every day, so the amount alone tells you very little.

What to check, in order

1. Is the beneficiary on the approved list in Section 4.6, and is the amount consistent with its history?
2. If the beneficiary is new, when was it added and by whom?
3. If bank details on an existing beneficiary changed, how was the change requested and was it verified out of band?
4. Was it initiated inside business hours by an authorised initiator, and was dual approval satisfied?
5. Are there several transfers that look designed to stay under a threshold?

Close if

- Established beneficiary, amount within its normal range, initiated in hours by an authorised person.
- A change or approval record documents it, with the approvals recorded.

Escalate if

- A new beneficiary receives a significant first payment, particularly outside business hours.
- Bank details on an existing beneficiary changed shortly before a scheduled payment and the request arrived by email.
- Multiple transfers sit just under a reporting or approval threshold.
- The initiating account is the subject of any other alert in the queue.

That last criterion is doing a lot of work

Wire fraud almost never appears as a single wire alert. It appears as a login alert, then a mailbox alert, then a wire alert, hours apart, in three different categories, each one individually explainable.

Cross-check every wire alert against the account names in the rest of your queue before you decide.

6. Day 1: The Alert Queue

Twenty alerts. Work them in any order. Record a decision and a justification for every one.

6.1 Failed login alerts

MTB-A01 FAILED LOGIN THRESHOLD Tue 08:41 EDT Severity: Low
 Account : cwalsh
 Source : 10.50.1.31
 Detail : 7 consecutive authentication failures 08:41:02 to 08:43:48,
 followed by a successful authentication at 08:44:11.
 MFA : Satisfied, approved push, first prompt.

MTB-A02 FAILED LOGIN THRESHOLD Tue 23:18 EDT Severity: Medium
 Accounts : 312 distinct accounts
 Source : 45.134.26.91 (external, geolocated to Netherlands)
 Detail : 1,284 authentication failures in an 11 minute window against the
 external webmail portal. Between 3 and 5 attempts per account.
 No successful authentications recorded.
 MFA : Not reached.

MTB-A03 FAILED LOGIN THRESHOLD Wed 02:00 EDT Severity: Low
 Account : svc-backup-agent
 Source : 10.50.2.25
 Detail : 4 authentication failures at 02:00:03. No success. This alert has
 fired at the same time every night for the past 14 nights.
 MFA : Not applicable, service account.

MTB-A04 FAILED LOGIN THRESHOLD Tue 23:47 EDT Severity: Medium
 Account : dkoenig
 Source : 10.50.1.77
 Detail : 9 authentication failures 23:47:19 to 23:51:02, followed by a
 successful authentication at 23:51:40. Account holds domain
 administrator privilege.
 MFA : Satisfied on the fourth push prompt.

MTB-A05 FAILED LOGIN THRESHOLD Wed 07:15 EDT Severity: Low
 Account : mnguyen
 Source : 198.51.100.22
 Detail : 3 authentication failures 07:15:44 to 07:16:20, followed by a
 successful authentication at 07:16:58 from a newly registered
 mobile device.
 MFA : Satisfied, approved push, first prompt.

6.2 Impossible travel alerts

MTB-B01 IMPOSSIBLE TRAVEL Wed 04:01 EDT Severity: High
 Account : rpatel
 Login 1 : Tue 03:14 EDT, 10.50.1.60, Pittsburgh US
 Login 2 : Wed 04:01 EDT, 102.89.44.7, Lagos NG
 Elapsed : 47 minutes
 Post-login: No sensitive system access recorded in the following hour.
 MFA : Satisfied on the sixth push prompt.

MTB-B02 IMPOSSIBLE TRAVEL Tue 10:20 EDT Severity: Medium
 Account : jhaddad
 Login 1 : Tue 06:20 EDT, 10.50.1.63, Pittsburgh US
 Login 2 : Tue 10:20 EDT, 72.44.19.203, Miami US
 Elapsed : 4 hours 0 minutes
 Post-login: Opened the commercial lending pipeline report. Normal for this user.
 MFA : Satisfied, approved push, first prompt.

MTB-B03 IMPOSSIBLE TRAVEL Wed 09:23 EDT Severity: Medium
 Account : lbrennan
 Login 1 : Wed 09:02 EDT, 10.50.1.71, Pittsburgh US
 Login 2 : Wed 09:23 EDT, 198.51.100.41, Columbus US
 Elapsed : 21 minutes
 Post-login: Routine access to data services tooling.
 MFA : Satisfied, approved push, first prompt.

MTB-B04 IMPOSSIBLE TRAVEL Tue 22:58 EDT Severity: High
 Account : tokafor
 Login 1 : Tue 21:50 EDT, 10.50.1.58, Pittsburgh US
 Login 2 : Tue 22:58 EDT, 89.36.216.140, Bucharest RO
 Elapsed : 1 hour 8 minutes
 Post-login: Accessed the wire transfer initiation system at 23:04.
 MFA : Satisfied on the ninth push prompt.

MTB-B05 IMPOSSIBLE TRAVEL Wed 09:15 EDT Severity: Medium
 Account : sgrigsby
 Login 1 : Wed 07:40 EDT, 10.50.1.66, Pittsburgh US
 Login 2 : Wed 09:15 EDT, 99.240.18.66, Toronto CA
 Elapsed : 1 hour 35 minutes
 Post-login: Accessed the internal audit workpaper repository.
 MFA : Satisfied, approved push, first prompt.
 Note : No travel request on file. Audit staff travel is booked departmentally and is not always recorded centrally.

6.3 Business email compromise indicator alerts

MTB-C01 MAILBOX RULE CREATED Tue 23:09 EDT Severity: High
 Mailbox : tokafor
 Created by: tokafor, from 89.36.216.140
 Rule name : ..
 Rule logic: IF subject or body contains any of: wire, invoice, payment, beneficiary, confirm, approval

THEN move to folder 'RSS Subscriptions' AND mark as read

MTB-C02 LOOKALIKE DOMAIN Tue 19:44 EDT Severity: High
 From : t.okafor@meridian-trust.com
 To : tokafor@meridiantrust.com
 Subject : Treasury portal - action required before cut-off
 Detail : Sender domain registered 9 days ago. SPF: fail. DKIM: none.
 DMARC: none. Message contained a link to a credential entry page
 styled as the Meridian Trust treasury portal.
 Gateway : DELIVERED to inbox. Not quarantined.

MTB-C03 EXTERNAL AUTO-REPLY Tue 07:02 EDT Severity: Low
 Mailbox : jhaddad
 Detail : Out-of-office auto-reply enabled for external senders. Text reads:
 'I am at a conference in Miami until Friday. For urgent lending
 matters contact Commercial Lending duty desk.'
 Created by: jhaddad, from 10.50.1.63, Mon 16:50 EDT

MTB-C04 EXTERNAL FORWARDING RULE Tue 14:31 EDT Severity: Medium
 Mailbox : bfoster
 Created by: bfoster, from 10.50.1.80
 Rule name : 'Copy to personal'
 Rule logic: IF sender is in 'Marketing Vendors' contact group
 THEN forward a copy to b.foster.home@examplemail.com
 Detail : Rule created during business hours from the user's own workstation.
 No attempt to hide the rule. User has not been contacted.

MTB-C05 DISPLAY NAME IMPERSONATION Wed 06:12 EDT Severity: Medium
 From : diane.kessler.cfo@examplemail.com
 Display : 'Diane Kessler'
 To : 1 recipient (Accounts Payable shared mailbox)
 Subject : Quick favour - are you at your desk?
 Detail : No links, no attachments. Generic text. Classic gift-card pretext.
 Gateway : QUARANTINED before delivery. Recipient never saw it.
 Reported independently by a second employee who received a
 similar message at a previous employer.

6.4 Suspicious wire transfer alerts

MTB-D01 NEW BENEFICIARY, HIGH VALUE Tue 23:12 EDT Severity: High
 Initiated : tokafor
 Amount : 47,500.00 USD
 Beneficiary: Sorana Trading SRL
 Bank : Banca Transilvania, Romania
 Detail : Beneficiary added to the payee list at 23:06 the same evening.
 First payment to this beneficiary. Initiated 6 hours after the
 17:00 wire cut-off. Dual approval: NOT YET OBTAINED, queued.

MTB-D02 RECURRING HIGH VALUE Tue 09:30 EDT Severity: Low
 Initiated : tokafor

Amount : 129,880.00 USD
 Beneficiary: ADP Payroll Services
 Bank : Wells Fargo, US
 Detail : Fortnightly payroll run. Amount within the normal range for this beneficiary. Dual approval obtained from Treasury lead.

MTB-D03 THRESHOLD PATTERN Tue 11:02 - 16:40 Severity: Medium
 Initiated : rharper (Accounts Payable)
 Detail : 5 outbound transfers in one day, all to different beneficiaries, none previously used:
 11:02 9,850.00 USD 13:15 9,700.00 USD
 14:28 9,900.00 USD 15:51 9,750.00 USD
 16:40 9,800.00 USD
 Total 49,000.00 USD. No single transfer reached the 25,000 USD dual-approval threshold, so none required a second approver.

MTB-D04 BENEFICIARY DETAILS CHANGED Mon 15:20 EDT Severity: High
 Beneficiary: Castlereagh Supply Co
 Changed by : rharper (Accounts Payable)
 Detail : Account number and receiving bank changed from PNC Bank US to an account at a different institution. Change requested by an email from the vendor's accounts contact. No callback to a previously known telephone number was recorded.
 Scheduled payment of 88,400.00 USD due Thursday.

MTB-D05 HIGH VALUE Tue 14:05 EDT Severity: Low
 Initiated : tokafor
 Amount : 250,000.00 USD
 Beneficiary: Harrow Logistics Ltd
 Bank : Barclays, UK
 Detail : Established beneficiary. Amount at the top of its normal range but within it. Approval reference CR-2291 recorded. Dual approval obtained from the CFO and the Treasury lead.

6.5 Step by step

1. Read Sections 4 and 5 before you touch the queue. Most wrong answers on this lab come from skipping the reference data.
2. Work each alert. Identify the type, apply the runbook, decide CLOSE or ESCALATE, and record your justification and confidence.
3. Do the arithmetic on every impossible-travel alert yourself. Do not take the alert's word for it, and do not take the word 'impossible' in the alert title as a finding.
4. Check every address in the queue against Section 4.3 before you treat it as external.
5. Fill in the Linked alerts column as you go. When you have finished all twenty, sort your log by account name and look again.
6. Count your escalations. If you escalated more than about half the queue, re-read the ones you are least sure about. If you escalated fewer than five, do the same.

- Identify the two alerts you found genuinely ambiguous, and write a paragraph on each explaining what would settle it and who you would ask.

One account appears in more than one category

Sort your triage log by account name before you call the queue finished. That single step is the difference between twenty individual alerts and one incident.

If you find it, say so explicitly in your escalation note. That note is what Day 2 starts from.

Alert ID	Host / Account	Type	Decision	Confidence	Justification	Runbook step	Linked alerts
MTB-A01	cwalsh	Failed Login	CLOSE	HIGH	7 failed logins within a 2:46 minute timeframe. Internal user Carla Walsh attempts login within business hours and approves MFA.	Close if: “Fewer than 10 failures on a single ordinary account from that user's own workstation or the VPN range, followed by a success, inside business hours.”	
MTB-A02	N/A	Failed Login	ESCALATE	HIGH	1,284 authentication failures on 312 distinct accounts in an 11 minute window, with no successful authentications recorded. Source IP from Netherlands at 23:18 EDT.	Escalate if: “Failures span multiple accounts from a single source. This is password spraying and the count does not need to be high.” & “The source is external or unrecognised.”	
MTB-A03	svc-backup-agent	Failed Login	CLOSE	MEDIUM	Pattern of 4 authentication failures at 2:00:03 EDT for the past 14 nights. Part of the nightly backup job that authenticates from source 10.50.2.25 at 02:00 daily. Expected behavior, though I believe the backup should authenticate	Close if: “A service account failing from its expected source in its expected pattern, with a known open ticket.”	

					successfully.		
MTB-A0 4	dcoenig	Failed Login	ESCALATE	HIGH	9 authentication failures in a 3:43 minute span. Successful authentication to a Domain Admin account at 23:51:40 EDT, outside business hours and from an unapproved workstation (IP 10.50.1.77). MFA satisfied on 4th push. Employee Daniel Koenig is on leave all week, there should be no logins.	Escalate if: “The account is privileged and the attempt succeeded.” & “The activity falls outside the user's working pattern with no change record, and especially if the user is known to be away.”	
MTB-A0 5	mnguyen	Failed Login	CLOSE	HIGH	3 authentication failures in 36 seconds followed by a successful authentication at 07:16:58 EDT from a newly registered device that Minh Nguyen enrolled in MDM on Monday. MFA approved first push. IP within VPN range.	Close if: “Fewer than 10 failures on a single ordinary account from that user's own workstation or the VPN range, followed by a success, inside business hours.” & “A change record explains it, such as a password rotation or a new device enrolment.”	
MTB-B0 1	rpatel	Impossible Travel	ESCALATE	HIGH	Login 1 from Pittsburgh and login 2 from Lagos are 47 minutes apart. Pittsburgh and Lagos are 5,578 miles away; login not possible. MFA approved on 6th push.	Escalate if: “The required speed exceeds roughly 600 miles per hour. No person moves that fast.” & “The second location is somewhere the user has never logged in from and there is no travel record.”	

MTB-B0 2	jhaddad	Impossible Travel	CLOSE	HIGH	Login 1 from approved Pittsburgh workstation on Tue 06:20 EDT. Login 2 from approved VPN in Miami on Tue 10:20 EDT. MFA approved on 1st push. User approved for travel to Miami for conference Tue to Fri.	Close if: “The required speed is achievable by commercial flight and there is approved travel on record.” & “One of the two addresses belongs to the corporate VPN egress range.”	
MTB-B0 3	lbrennan	Impossible Travel	CLOSE	HIGH	Login 1 from approved Pittsburgh workstation on Wed 09:02 EDT. Login 2 from approved VPN in Columbus on Wed 09:23 EDT. MFA approved on 1st push. VPN traffic leaves Meridian Trust’s network in Columbus Ohio.	Close if: “One of the two addresses belongs to the corporate VPN egress range.”	
MTB-B0 4	tokafor	Impossible Travel	ESCALATE	HIGH	Login 1 from approved Pittsburgh workstation on Tue 21:50 EDT. Login 2 from unrecognized IP in Bucharest on Tue 22:58 EDT. MFA approved on 9th push. Access wire transfer initiation system at 23:04 EDT.	Escalate if: “The required speed exceeds roughly 600 miles per hour. No person moves that fast.” & “The second location is somewhere the user has never logged in from and there is no travel record.” & “The account performed sensitive actions immediately after the second login.”	Link 1
MTB-B0	sgrigsby	Impossible	ESCALATE	MEDIUM	Login 1 from	Escalate if: “The	

5		Travel			approved Pittsburgh workstation on Wed 07:40 EDT. Login 2 from unrecognized IP in Toronto on Wed 09:15 EDT. MFA approved on 1st push. Accessed the internal audit workpaper repository. Although plausible to travel, highly unlikely the user logged in on a corporate workstation in Pittsburgh, then traveled to Toronto and logged in from an unknown device. Unknown IP is very suspicious.	second location is somewhere the user has never logged in from and there is no travel record.”	
MTB-C01	tokafor	BEC	ESCALATE	HIGH	Rule made on Tue 23:09 from unrecognized IP to move emails that contain “wire, invoice, payment, beneficiary, confirm, approval” to folder “RSS Subscriptions” and mark as read. Rule made outside of business hours and looks for financial keywords.	Escalate if: “A rule moves, deletes or marks as read any message matching financial keywords. Treat this as compromise until proven otherwise.”	Link 1
MTB-C02	Tokafor (email)	BEC	ESCALATE	HIGH	Tokafor email (tokafor@meridianrust.com) received an email from a spoofed sender (t.okafor@meridian-trust.com) with an email subject	Escalate if: “A lookalike domain is involved, whether or not the message was delivered.” & “An executive or an authorised	Link 1

					“Treasury portal - action required before cut-off”. The sender domain was registered 9 days ago and the message contained a link to a credential entry page styled as the Meridian Trust treasury portal.	payment initiator is impersonated or targeted.”	
MTB-C03	Jhaddad (email)	BEC	CLOSE	HIGH	Auto-reply message created to inform senders he is at a conference in Miami until Friday. Created by Jhaddad from a verified workstation on Mon 16:50 EDT. Lines up with his Miami conference from Tuesday to Friday.	Close if: “The activity is an ordinary mail feature used ordinarily, such as an out-of-office reply set by a user who is genuinely on leave.”	
MTB-C04	Bfoster (email)	BEC	ESCALATE	MEDIUM	Rule created on Tue 14:31 EDT from verified workstation. Rule Logic: “IF sender is in 'Marketing Vendors' contact group THEN forward a copy to b.foster.home@examplemail.com ”. Forward email is outside of domain.	Escalate if: “A rule forwards mail to an address outside the bank.”	
MTB-C05	diane.kessler.cfo@examplemail.com	BEC	ESCALATE	MEDIUM	Message sent from “ diane.kessler.cfo@examplemail.com ” on Wed 06:12 EDT with subject: “Quick favour - are you at your desk?”. Email quarantined.	Escalate if: “A lookalike domain is involved, whether or not the message was delivered.”	

					Sent outside of business hours and email appears to be spoofed (I don't have a list of legitimate emails).		
MTB-D01	tokafor	Susp. Wire Transfers	ESCALATE	HIGH	Tokafor initiated a \$47,500 USD wire transfer to Sorana Trading SRL (Bank: Banca Transilvania, Romania) on Tue 23:12 EDT, 6 hours after the wire cut-off. Beneficiary added to the payee list at 23:06 the same evening.	Escalate if: "A new beneficiary receives a significant first payment, particularly outside business hours." & "The initiating account is the subject of any other alert in the queue."	Link 1
MTB-D02	tokafor	Susp. Wire Transfers	CLOSE	HIGH	Tokafor initiated \$129,880 USD wire transfer to ADP on Tue 09:30 EDT. Fortnightly payroll run. Amount within normal range for beneficiary (ADP Payroll Services) and dual approval obtained from Treasury lead.	Close if: "Established beneficiary, amount within its normal range, initiated in hours by an authorised person."	
MTB-D03	rharper	Susp. Wire Transfers	ESCALATE	HIGH	Rharper sent 5 outbound transfers within a approx. 5 hour window totalling \$49,000 USD (11:02 9,850.00 USD, 13:15 9,700.00 USD, 14:28 9,900.00 USD, 15:51 9,750.00 USD, 16:40 9,800.00 USD). All different beneficiaries, none previously used. All under the	Escalate if: "A new beneficiary receives a significant first payment, particularly outside business hours." & "Multiple transfers sit just under a reporting or approval threshold." & "The initiating account is the subject of any other alert in the	Link 2

					\$25,000 dual-approval threshold.	queue.”	
MTB-D04	rharper	Susp. Wire Transfers	ESCALATE	HIGH	On Mon 15:20 EDT, rharper changed beneficiary details to a different account number and institution (Castlereagh Supply Co). “Change requested by an email from the vendor's accounts contact. No callback to a previously known telephone number was recorded. Scheduled payment of 88,400.00 USD due Thursday.”	Escalate if: “Bank details on an existing beneficiary changed shortly before a scheduled payment and the request arrived by email.” & “The initiating account is the subject of any other alert in the queue.”	Link 2
MTB-D05	tokafor	Susp. Wire Transfers	CLOSE	HIGH	Tokafor established beneficiary Harrow Logistics LTD (Bank: Barclays, UK) for wire amount of \$250,000 USD. “Amount at the top of its normal range but within it. Approval reference CR-2291 recorded. Dual approval obtained from the CFO and the Treasury lead.”	Close if: “Established beneficiary, amount within its normal range, initiated in hours by an authorised person.” & “A change or approval record documents it, with the approvals recorded.”	

7. Day 1: Writing Escalation Notes

Everything you escalate needs a note. The note is the entire handover. Tier 2 will act on what you wrote, not on what you meant, and on a busy shift they will not come and ask you.

7.1 The format

Field	What goes in it
Alert ID and type	Which alert, and what kind
What was observed	The facts only. Accounts, addresses, times, amounts. No interpretation yet.
Why it is suspicious	Your reasoning, and the runbook criterion you applied
What Tier 2 should look at first	The single most useful next step. Be specific.
Related alerts	Any other alert ID you think belongs with this one
Urgency	Your assessment, and why. Money in motion beats everything else.

7.2 A worked escalation note

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ALERT      : MTB-A02, failed login threshold
OBSERVED   : 1,284 authentication failures against the external webmail portal
              between 23:18 and 23:29 Tuesday, from 45.134.26.91, spread across
              312 distinct accounts at 3 to 5 attempts each. No successes.
SUSPICIOUS : The low per-account attempt count combined with the very high
              account count is the signature of password spraying rather than
              brute force. The attempt count per account sits deliberately below
              our lockout threshold. Failed Login Runbook escalation criterion 1.
FIRST STEP : Confirm whether any of the 312 accounts authenticated successfully
              from any source in the following 24 hours. The absence of success
              during the spray does not mean the attacker learned nothing.
RELATED    : None identified.
URGENCY    : Medium. No confirmed compromise, but the source is still active
              and the account list suggests prior enumeration of our directory.
  
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What makes that note useful

It separates what was seen from what it means, so Tier 2 can disagree with the interpretation without having to re-derive the facts.

The first step is a specific question with a yes or no answer, not 'investigate further'.

It names the runbook criterion, so the decision can be reviewed against procedure months later.

It says what the absence of evidence does not prove. That sentence is what separates a Tier 1 analyst from a ticket forwarder.

7.3 Step by step

1. Write a note in the Section 7.1 format for every alert you escalated.
2. For your highest-urgency escalation, write the first step as a question that can be answered yes or no.

3. For every alert you closed, write one or two sentences of justification in your triage log. A closure with no reasoning is itself a finding your shift lead will raise.
4. Pick the three closures you are least confident about and add a sentence to each saying what would change your mind.
 - MTB-A03
 - Confirmation that the scheduled job is supposed to fail 4 times without succeeding.
 - MTB-B02
 - Confirmation Joseph Haddad logged into the Pittsburgh workstation at 06:20 EDT, 40 minutes before business hours.
 - MTB-B03
 - If Columbus Ohio was not the destination for VPN traffic to leave, I would be concerned.

Escalation Notes

ALERT: MTB-A02, failed login threshold

OBSERVED: 1,284 authentication failures against the external webmail portal between 23:18 and 23:29, spread across 312 distinct accounts. Source IP located in the Netherlands. No successful authentications recorded.

SUSPICIOUS: Failures span multiple accounts from a single source. The source IP is external and unrecognized. Escalation criteria met.

FIRST STEP: Confirm whether any of the 312 accounts authenticated successfully from any source in the following 24 hours. The absence of success during the spray does not mean the attacker learned nothing.

RELATED: None identified.

URGENCY: Medium. No confirmed compromise, but the source is external, unrecognized, and the account list suggests prior enumeration of our directory.

ALERT: MTB-A04, failed login threshold

OBSERVED: 9 authentication failures in a 3:43 minute span against account dkoenig, followed by a successful authentication to a Domain Admin account at 23:51:40 EDT. The login occurred outside business hours, from an unapproved workstation (IP 10.50.1.77), with MFA satisfied on the 4th push attempt.

SUSPICIOUS: The account is privileged and the attempt succeeded. The activity falls outside the user's working pattern, with no change record on file and Employee Daniel Koenig is on leave all week, meaning there should be no logins from this account at all. Escalation criteria met on both counts.

FIRST STEP: Contact Daniel Koenig through an out-of-band channel (not email) to confirm whether he initiated this login. Given he is on leave and reportedly not working, treat this as a likely compromised credential until confirmed otherwise, and consider suspending the account pending confirmation.

RELATED: None identified.

URGENCY: High. Successful privileged access from an unapproved device, outside business hours, by a user believed to be away, combined with a repeated MFA push (possible push-fatigue attack), represents a strong compromise indicator.

ALERT: MTB-B01, impossible travel

OBSERVED: Login 1 for account rpatel originated from Pittsburgh; login 2 originated from Lagos, 47 minutes later. The two locations are 5,578 miles apart. MFA was approved on the 6th push attempt.

SUSPICIOUS: The implied travel speed is roughly 7,120 mph, far beyond the ~600 mph threshold that flags physically impossible travel; no person moves that fast. Lagos also has no travel record on file for this user, meaning there's no prior login history from that location. Both escalation criteria are met.

FIRST STEP: Contact rpatel through an out-of-band channel to confirm which login (if either) was legitimate. Treat the Lagos session as the likely attacker foothold given the impossible-travel math and the lack of any travel record, and consider suspending the account and revoking active sessions pending confirmation.

RELATED: None identified.

URGENCY: High. Impossible travel combined with an unrecorded/new location and a 6th-push MFA approval (a strong push-fatigue indicator, consistent with MTB-A04) suggests active account compromise rather than a benign explanation like VPN or travel.

ALERT: MTB-B04, impossible travel

OBSERVED: Login 1 for account tokafor originated from an approved Pittsburgh workstation Tuesday 21:50 EDT. Login 2 originated from an unrecognized IP in Bucharest, 68 minutes later at 22:58 EDT. MFA was approved on the 9th push attempt. The account then accessed the wire transfer initiation system at 23:04 EDT.

SUSPICIOUS: The implied travel speed is roughly 4,700 mph, far beyond the ~600 mph physical-travel threshold. Bucharest has no prior travel record for this user. The account also performed a sensitive action, initiating access to the wire transfer system, immediately after the second login. All three escalation criteria are met.

FIRST STEP: Treat this as active compromise, not just suspicious login activity, given the wire-transfer-system access. Suspend the account and revoke active sessions immediately, then contact tokafor through an out-of-band channel to confirm legitimacy. Loop in whoever owns wire transfer approvals now, to check whether any transfer was initiated or completed from this session.

RELATED: Alert MTB-C01, MTB-C02, and MTB-D01

URGENCY: High. This is impossible travel, a 9th-push MFA approval (the most aggressive fatigue signal seen so far), and touching a financial system, all within a 14-minute span.

ALERT: MTB-B05, impossible travel

OBSERVED: Login 1 for account sgrigsby originated from an approved Pittsburgh workstation Wednesday 07:40 EDT. Login 2 originated from an unrecognized IP in Toronto, 1 hour 35 minutes later at 09:15 EDT. MFA was approved on the 1st push. The account accessed the internal audit workpaper repository.

SUSPICIOUS: Pittsburgh to Toronto is physically travelable in this window, so this doesn't meet the same speed-based impossibility as prior alerts, but it's highly unlikely the user drove/flew from a corporate workstation login straight to a second login from an unrecognized device in another city with no travel record on file. Escalation criteria met on the unrecognized-location/no-travel-record basis.

FIRST STEP: Contact sgrigsby through an out-of-band channel to confirm whether they traveled to Toronto and logged in from a personal or unknown device. Given access to the audit workpaper repository, also confirm with the audit team whether anything sensitive was viewed or exported during the Toronto session.

RELATED: None identified.

URGENCY: Medium. This is plausible travel rather than physically impossible, and MFA was approved cleanly on the 1st push with no fatigue pattern. Still warrants confirmation given the unrecognized device and access to audit materials, but the compromise signal is weaker.

ALERT: MTB-C01, business email compromise (BEC)

OBSERVED: A mailbox rule was created for account tokafor on Tuesday at 23:09 from an unrecognized IP. The rule moves emails containing "wire," "invoice," "payment," "beneficiary," "confirm," or "approval" to a folder named "RSS Subscriptions" and marks them as read. The rule was created outside business hours.

SUSPICIOUS: A rule that moves, deletes, or marks as read any message matching financial keywords is to be treated as compromise until proven otherwise. This rule matches that pattern exactly; it's a classic mailbox-hiding technique used to intercept financial correspondence (e.g., wire fraud, invoice fraud) while suppressing the alerts the legitimate user would normally see. The disguised folder name ("RSS Subscriptions") indicates deliberate concealment, not user error. Escalation criteria met.

FIRST STEP: Treat this as confirmed compromise per the escalation rule. Immediately disable the mailbox rule, suspend the account, and revoke active sessions. Search the "RSS Subscriptions" folder and mailbox audit log for any financial correspondence that may have already been intercepted or actioned, and loop in whoever owns wire transfer approvals to check for fraudulent payment activity tied to this account.

RELATED: Alert MTB-B04, MTB-C02, and MTB-D01

URGENCY: High. This is an active BEC/wire fraud pattern building directly on MTB-B04 — same account, same session timeframe, now with intercept capability for financial communications. Should be escalated as a single high-severity incident, not two isolated alerts.

ALERT: MTB-C02, business email compromise (BEC)

OBSERVED: The Tokafor email account (tokafor@meridiantrust.com) received a message from a spoofed sender, t.okafor@meridian-trust.com, subject line "Treasury portal - action required before cut-off." The sender domain was registered 9 days ago and contained a link to a credential entry page styled to impersonate the Meridian Trust treasury portal.

SUSPICIOUS: A lookalike domain is involved (meridian-trust.com vs. the legitimate meridiantrust.com), which meets the escalation criteria regardless of whether the message was actually delivered. The payment initiator (Tokafor) is being impersonated or directly targeted, and the recently-registered domain plus fake credential-harvesting portal are hallmarks of a targeted phishing attempt rather than generic spam. Escalation criteria met on both counts.

FIRST STEP: Confirm delivery status and whether Tokafor (or anyone else) interacted with the link or entered credentials. Block the lookalike domain at the email gateway/firewall, and if credentials were entered, treat this as confirmed compromise. Reset the account, revoke sessions, and follow the same containment steps as MTB-B04/MTB-C01.

RELATED: Alert MTB-B04, MTB-C01, and MTB-D01

URGENCY: High. This appears to be the root cause of the Tokafor incident chain already in progress. Confirming whether credentials were actually entered on the fake portal is the key open question and should be prioritized.

ALERT: MTB-C04, business email compromise (BEC)

OBSERVED: A mailbox rule was created for account Bfoster on Tuesday 14:31 EDT from a verified workstation. Rule logic: if sender is in the "Marketing Vendors" contact group, forward a copy to b.foster.home@examplemail.com. The forwarding address is outside the bank's domain.

SUSPICIOUS: A rule that forwards mail to an address outside the bank meets the escalation criteria as written. That said, this instance has notably weaker compromise indicators than the other C-series alerts: the rule was created from a verified (not unrecognized) workstation, during business hours, the forwarding target appears to be a personal address matching the employee's own name ("b.foster.home"), and it's scoped narrowly to one contact group rather than financial keywords. This pattern is as consistent with an employee forwarding vendor mail to a personal account (against policy, but not necessarily malicious) as it is with compromise.

FIRST STEP: Contact Bfoster directly to confirm whether they created this rule intentionally and why. Review what "Marketing Vendors" mail has already been forwarded to the external address to assess data exposure, and confirm the rule against acceptable-use/data-handling policy regardless of intent.

RELATED: None identified.

URGENCY: Medium, per the table. Given the verified workstation, business-hours creation, and personal-looking destination address, this reads more like a likely policy violation than an active compromise, but confirm with the user before downgrading it, since an attacker could also use a personal-looking address as a low-visibility exfil channel.

ALERT: MTB-C05, business email compromise (BEC)

OBSERVED: A message was sent from diane.kessler.cfo@examplemail.com on Wednesday 06:12 EDT, subject "Quick favour - are you at your desk?" Sent outside business hours. The email was quarantined.

SUSPICIOUS: A lookalike domain is involved, which meets the escalation criteria regardless of whether the message was actually delivered, and in this case it was quarantined, not delivered, but the rule still applies. This follows the classic "urgent favor" CFO-impersonation pretext used to open a BEC conversation before requesting a wire or gift cards. Sent outside business hours adds to the suspicion.

FIRST STEP: Confirm the legitimate domain/email format for the actual CFO to verify this is spoofed, and check whether it was sent to other employees besides the one it landed with. Since it's quarantined, no user interaction has occurred yet. Block the sending domain and address at the gateway to prevent any related follow-up messages from landing.

RELATED: None identified.

URGENCY: Low. No user interaction occurred since the message was quarantined, so there's no confirmed exposure, but the CFO-impersonation pretext and off-hours timing suggest this may be an opening move in a BEC attempt, worth watching for related messages to other staff.

HIGHEST PRIORITY!!!

ALERT: MTB-D01, suspicious wire transfer

OBSERVED: Tokafor initiated a \$47,500 USD wire transfer to Sorana Trading SRL (Bank: Banca Transilvania, Romania) on Tuesday 23:12 EDT, 6 hours after the standard wire cut-off. The beneficiary was added to the payee list at 23:06 that same evening, 6 minutes before the transfer was initiated.

SUSPICIOUS: A new beneficiary received a significant first payment, particularly outside business hours. The beneficiary was added and paid within 6 minutes, well outside business hours, and past cut-off. The initiating account is also the subject of other alerts already in the queue (Tokafor: MTB-B04, MTB-C01, MTB-C02), the second escalation criterion. Both criteria met.

FIRST STEP: Treat as confirmed fraud given the account's existing compromise chain. Contact the bank/wire operations team immediately to attempt a recall or hold on the \$47,500 transfer before it settles internationally. Suspend the Tokafor account and revoke sessions if not already done under MTB-B04/C01.

RELATED: MTB-B04, MTB-C01, and MTB-C02.

URGENCY: High and time-critical. Unlike the earlier alerts in this chain, this one involves money actually leaving the bank. Wire recall windows are short, so this should be flagged for immediate action ahead of writeup formatting.

ALERT: MTB-D03, suspicious wire transfer

OBSERVED: Rharper sent 5 outbound wire transfers within approximately a 5-hour window, totaling \$49,000 USD to 5 different, previously unused beneficiaries: 11:02 (\$9,850.00), 13:15 (\$9,700.00), 14:28 (\$9,900.00), 15:51 (\$9,750.00), and 16:40 (\$9,800.00). All transfers fell just under the reporting/approval threshold.

SUSPICIOUS: Multiple transfers sitting just under a reporting or approval threshold is a named escalation criterion here, and this pattern matches it precisely. Five transfers, each individually below the threshold, that sum to nearly \$49,000. New beneficiaries receiving significant first payments is a second criterion met across all five. This is a classic structuring pattern (deliberately splitting a transfer to avoid triggering approval controls).

FIRST STEP: Attempt to place a hold/recall on all 5 transfers immediately given the structuring pattern and time-sensitivity. Suspend rharper's wire transfer privileges pending investigation, and contact rharper through an out-of-band channel to confirm intent.

RELATED: MTB-D04

URGENCY: High. Structuring below a reporting threshold across 5 transfers in 5 hours is a strong deliberate-evasion signal, and this involves money that may already be leaving the bank, making this alert time-sensitive for recall.

ALERT: MTB-D04, suspicious wire transfer

OBSERVED: On Monday 15:20 EDT, rharper changed beneficiary details on an existing vendor to a different account number and institution (Castlereagh Supply Co.). The change was requested by an email from the vendor's accounts contact. No callback was made to a previously known telephone number to verify the request. A scheduled payment of \$88,400.00 USD to this beneficiary is due Thursday.

SUSPICIOUS: Bank details on an existing beneficiary changed shortly before a scheduled payment, with the request arriving by email, is the classic vendor email compromise (VEC) / invoice fraud pattern (an attacker impersonates a known vendor to redirect a legitimate, expected payment). The lack of a callback to a previously known number to verify the change is the critical control failure here. The initiating account (rharper) is also now the subject of another alert in the queue (MTB-D03), the second escalation criterion. Both criteria met.

FIRST STEP: Before Thursday's \$88,400 payment executes, independently verify the beneficiary change by calling the vendor's previously known contact number (not any number in the change request email). Place a hold on the scheduled payment until verified. Given the D03 link, treat this alongside that alert rather than in isolation.

RELATED: MTB-D03

URGENCY: High, and time-critical ahead of Thursday's scheduled payment